CIVL1113 Engineering
Mechanics and Materials
May 2024 Examination
Comments on student performance
Mechanics part

General: Mechanics part

- The performance of students varies greatly, and the average performance was fine. In general, most of students were able to solve the problem and showed good understanding on the course content in mechanics part. The mechanics part exam is slightly easier than the homework. All the techniques needed to solve the problems have been covered in lectures.
- The pass ratio and average score for this semester course has increased and I am proud to see the increase.
- A problem of some students is the lack of understanding of internal forces in a structure, which is related to inability to construct bending moment diagram (BMD) and shear force diagram (SFD) diagram correctly. Looking at BMDs and SFDs for typical cases will help.
- Some students did not have the ability to spot problems from unusual results, e.g., convert between different units, bending moment diagram should be zero at pin joints unless external moment applied, work out reaction forces correctly, and to follow up with further checking.

Question 1

- The performance in this question was fine. The performance varied.
- In Question 1(a), the performance was generally fine but there are still some students who did not work out the results clearly. Some students directly draw wrong answer without detailed steps in calculating the reaction forces.
- Question 1(b), students should be able to understand that the beam is very similar to a simply supported beam and the column is also very similar to a cantilever column. Not all students follow the steps to obtain the results. Some students failed to obtain the reaction forces correctly.
- Some students presented BMD and SFD of wrong shapes, suggesting inadequate understanding of the concepts.
- Some students showed inadequate understanding of the sign conventions in the construction of BMD and SFD. Plotting the BMD on the tension side always presents the absolute values.

Question 2

- The overall performance in this question varied between students. The overall performance in this question was fine.
- For Q2a, some students had some difficulty in converting the unit systems in Q2(a). Some students did not use the correct equation to obtain the shear stress correctly. It is fine to use the precise equation or use the simplified method (divide shear force by web area for H-shaped section). Some students forget to check the intersection and some students forget to check top fiber of the section.
- Some students used a wrong equation for Q2(b) and some students were unable to construct the Mohr circle correctly. Only a relatively less students obtained all results correctly.
- Some students did not well use the strength criteria to check the safety as requested by the exam paper. I am very satisfied that many students were able to solve the von-Mises stress checking problem using the equation recommended by homework 4 recommended answer. I am very proud of you for self-motivated learning shown by the answers.

Question 3

- The overall performance in this question was fine.
- Part (a) is a relatively simple torsion problem. Some students did not use the equations correctly to find out the shear stress in the member. Some students were unable to solve the unit convert issues.
- Part (b) is buckling issue problem. Some students did not work out the effective length correctly. Some students did not consider out-of-plane buckling, which was clearly stated in the exam paper that need to consider. Some students did not work out the stress at buckling load.